

Radiation from Moving Charges

Refs: Weidemann Ch. 23, 24, 25

We've seen: $P_{\text{rad}} \propto \frac{E^4}{p^2}$

→ Limiting factor for colliders

→ Radiation has all sorts of useful properties as well!

Start by looking at conservation of Energy.

Imagine an e^- moving in vacuum

Energy $\rightarrow \gamma$ Momentum $\rightarrow \beta\gamma$

$$E = \gamma mc^2$$

$$p = \gamma \beta mc$$

From relativity: $E^2 = p^2 c^2 + m^2 c^4$

$$\gamma^2 m^2 c^4 = (\beta\gamma)^2 m^2 c^4 + m^2 c^4$$

$$\rightarrow \boxed{\gamma = \frac{1}{\sqrt{1 - (\beta)^2}}} \Rightarrow \text{Dispersion Relation for a particle.}$$

For a photon:

$$E_p = \hbar \omega$$

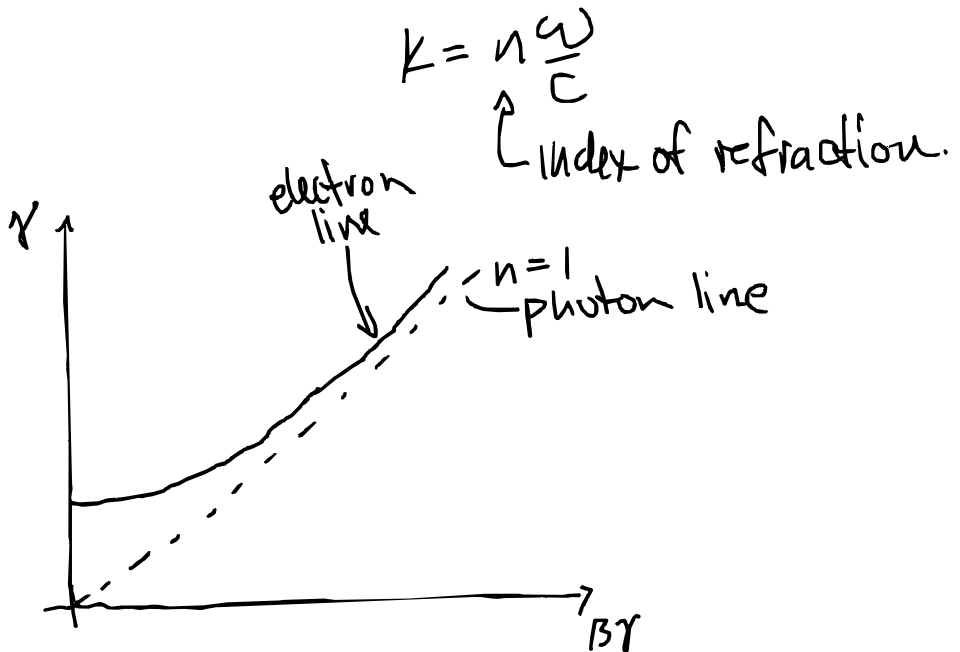
$$p_p = \hbar k_p$$

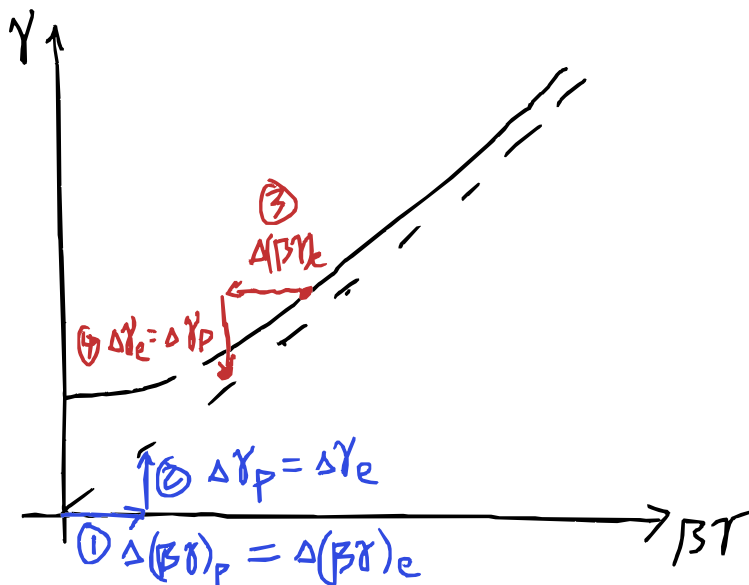
Normalize both with e^- mass

$$\gamma_p = \frac{E_p}{mc^2} = \frac{\hbar \omega}{mc^2}$$

$$(\beta\gamma)_p = \frac{\hbar k_p}{mc}$$

$$\rightarrow \frac{\gamma_p}{(\beta\gamma)_p} = \frac{\omega}{k_p c} = \frac{1}{n} < 1$$





Can a free particle radiate in a vacuum?

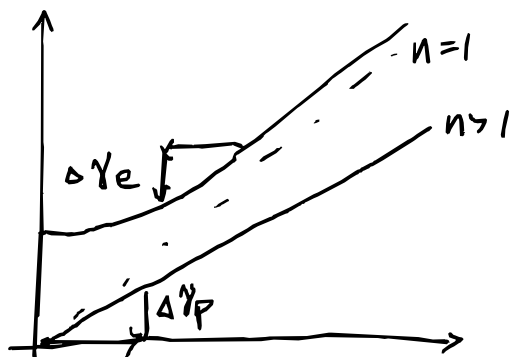
- ① Create a photon with mtn $\Delta(\beta\gamma)_p$
- ② That photon has energy $\Delta\gamma_p$
- ③ Electron must lose mtn $\Delta(\beta\gamma)_e = \Delta(\beta\gamma)_p$
- ④ " " " Energy $\Delta\gamma_e = \Delta\gamma_p$

But!

This pushes the e^- off of the dispersion line

→ Emission of a photon violates conservation of energy.

What if we are not in vacuum



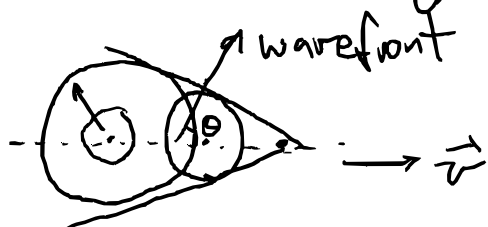
$$\Delta(\gamma\beta)_p = \Delta(\gamma\beta)_e$$

$$\Delta\gamma_p = \Delta\gamma_e$$

$\Rightarrow$ Cherenkov radiation.

Because there cannot be a change to $P_{\perp}$
 $\rightarrow$ emission must be azimuthally symmetric.

Imagine the fields expanding from a moving charge:



$$\rightarrow n/\beta \cos \theta = 1$$

EM Radiation

Poynting vector $\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$

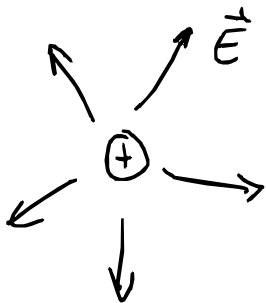
→ describes the propagation of energy in EM fields

∴ can only have radiation when

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \neq 0$$

Coulomb Regime:

→ static charge.



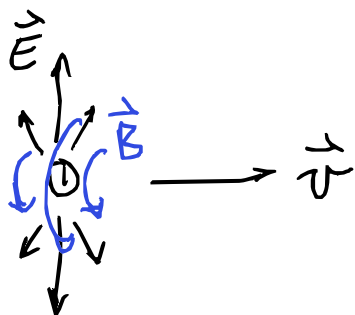
$$\vec{B} = 0$$

$$\vec{E} \times \vec{B} = 0$$

⇒ No radiation

what if the charge is moving at const. speed?

→ Do a Lorentz transformation of the fields



$$\Rightarrow \vec{E} \times \vec{B} \neq 0$$

E fields compressed

B field circulates the direction of motion

Do we have radiation?

fields drop off quickly w. distance.

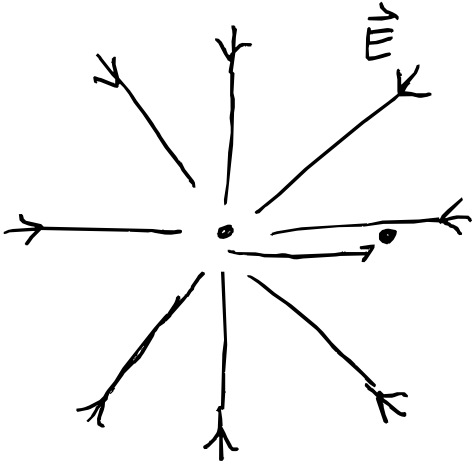
Energy is confined to the particle.

→ Not particularly interesting in this class.

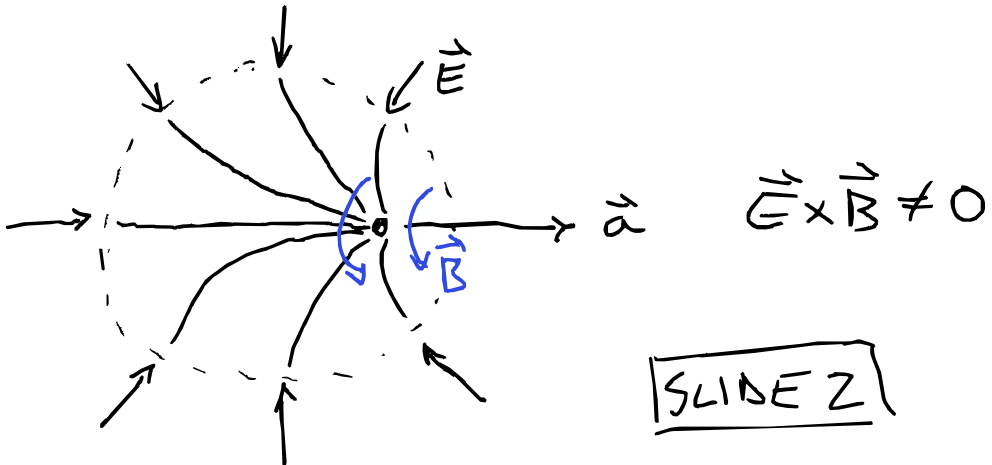
Radiation Regime

Let's look at an e^- starting at rest
 $\rightarrow$ accelerating

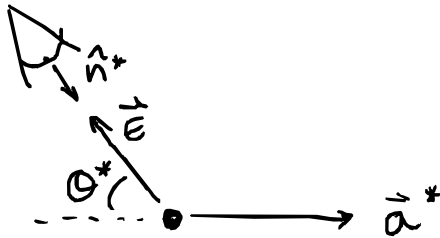
At rest, fields:



After a time Δt , and acceleration a



For an observer looking at the charge from an angle θ^* , unit vector $\hat{u}^*$



For $\vec{a}^*$ in direction of motion

$$\text{Ansatz: } \vec{E}_{\parallel}^* = -\frac{1}{4\pi\epsilon_0} \frac{e\vec{a}^*}{c^2 R} \sin\theta^*$$

(distance from charge)

$$\Rightarrow \vec{S} = \frac{1}{c\mu_0} E_{\parallel}^{*2} \hat{u}^*$$

Radiation dist. is $\propto \sin^2\theta^*$

→ donut distribution.

see Weidemann Fig 23.5

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Acceleration transverse to motion

→ Distortion of fields creates primarily transverse or radial field components

$$\vec{E}_{\perp}^* = -\frac{M_0 e \vec{a}^*}{4\pi R} \cos\theta^*$$

$$\Rightarrow \vec{S} = \frac{1}{c M_0} \vec{E}_{\perp}^* \hat{n}^*$$

See Fig 23.6

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→ v in z

acceleration in x

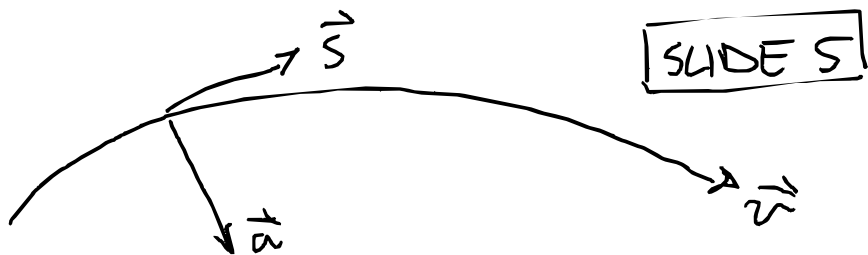
⇒ transverse $\vec{E}$

Pointing vector in $\vec{z}$

Synchrotron Radiation

Weidemann Ch 25

We just showed that a particle accelerating in a ring will radiate in the direction of motion



Let's look at the form of this radiation field

Start with the vector + scalar potentials

$$\vec{B} = \vec{\nabla} \times \vec{A} \quad \vec{E} = -\frac{d\vec{A}}{dt} - \nabla\phi$$

From these we get the wave eq's:

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{d^2 \vec{A}}{dt^2} = \frac{1}{\epsilon_0} \rho \vec{B}$$

$$\nabla^2 \phi - \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = -\frac{1}{\epsilon_0} \rho$$

Solutions to the wave equations are:

$$\vec{A}(t) = \frac{\mu_0}{4\pi} \int \frac{\vec{v} \rho(x,y,z)}{R} \Big|_{t_r} dx dy dz$$

$$\phi(t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(x,y,z)}{R} \Big|_{t_r} dx dy dz$$

Since things are relativistic

→ use retarded time $t_r = t - \frac{1}{c} R(t_r)$

with R = distance from observer to the charge, at time t_r

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For a point charge, moving w. $\vec{\beta} = \frac{\vec{v}}{c}$

→ Retarded potentials at time t ,
observation pt P ($\hat{n} \rightarrow$ normal vec. to observer)

$$\vec{A}(P,t) = \frac{1}{4\pi c\epsilon_0} \frac{q}{R} \frac{\vec{\beta}}{1 + \hat{n} \cdot \vec{\beta}} \Big|_{t_r}$$

$$\phi(P,t) = \frac{1}{4\pi\epsilon_0} \frac{q}{R} \frac{1}{1 + \hat{n} \cdot \vec{\beta}} \Big|_{t_r}$$

⇒ Liénard - Wiechert potentials

Using these potentials $\Rightarrow \vec{E}$ equation
(see Weidemann sec. 25.1)

$$4\pi\epsilon_0 \frac{\vec{E}}{q} = \underbrace{\frac{1-\beta^2}{r^3} (\vec{R} + R\vec{\beta})}_{\propto \frac{1}{r} \text{ when } \beta \rightarrow 0 \text{ Coulomb Field}} + \underbrace{\frac{1}{cr^3} \left\{ \vec{R} \times \left[(\vec{R} + R\vec{\beta})_r \times \frac{d\vec{\beta}}{dt} \right] \right\}}_{\text{Radiation Regime}}$$

Coulomb term $\rightarrow$ ignore. Energy is close to charge. Not of interest.

Radiation term $\rightarrow$ notice the accel. term!

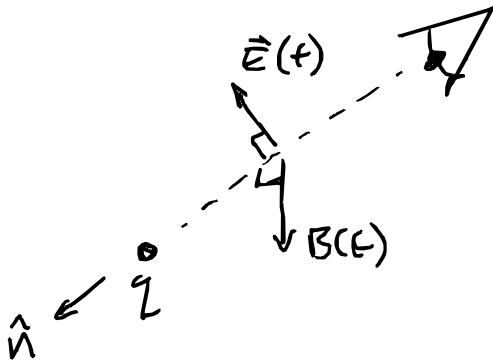
$$\Rightarrow \left| 4\pi\epsilon_0 \frac{\vec{E}(t)}{q} \right|_{\text{rad.}} = \frac{1}{cr^3} \left\{ \vec{R} \times \left[(\vec{R} + R\vec{\beta})_r \times \frac{d\vec{\beta}}{dt} \right] \right\}_r$$

Note $\rightarrow$ polarization is $\perp$ to $\vec{R}$
($\vec{R} \times \vec{\beta}$ terms)

Can similarly find $\vec{B}(t)$

→ We find that $\vec{E} \perp \vec{B}$

$$\vec{B} = \frac{1}{c} [\vec{E} \times \hat{n}]_r$$



Poynting vector: $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

$$= \epsilon_0 c [\vec{E} \times (\vec{E} \times \hat{n})]$$

$$\vec{S} = -\epsilon_0 c E_r^2 \hat{n} |_r$$

⇒ Flux is in the direction
of the observer

Total Radiation Power (sec 25.2)

Let's move to the ref. frame that moves w. the charged particle.

$$\vec{E}^*(t) = \frac{1}{4\pi\epsilon_0} \frac{e}{CR} \left(\hat{n} \times \left(\hat{n} \times \frac{d\vec{B}^*}{dt} \right) \right) \Big|_r$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ q \rightarrow e^- & & \vec{R} \rightarrow R\hat{n} \end{array}$$

The sync. radiated power per unit solid area

$$\text{is: } \frac{dP^*}{d\Omega} = -n \vec{S}^* R_r^2 = \epsilon_0 c \vec{E}^{*2} R^2 \Big|_r$$

$$= \left(\frac{r_e mc^2}{4\pi\epsilon_0} \frac{d\vec{B}^*}{dt} \right)^2 \sin^2 \theta_r$$

(classical e^- radius. SLIDE 7)

Note $\rightarrow$ this is our familiar donut distribution in the ref. frame of particle.

$$\text{Integrate: } P^* = \frac{2}{3} r_e mc^2 \left(\frac{d\vec{B}^*}{dt} \right)^2 \Big|_r$$

$\rightarrow$ Larmor Formula. (Accurate to within quantum limits)

Radiation Distributions

In the lab frame, we need to think about the particles motion.

→ For circular motion, i.e. $\vec{a} \perp \vec{v}$,

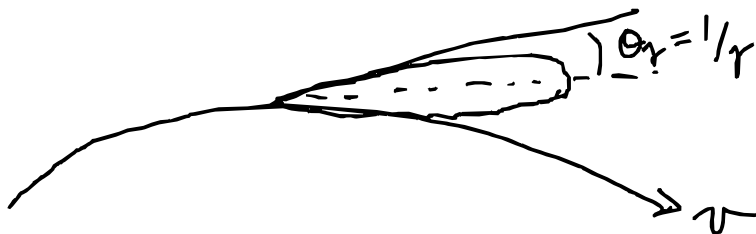
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$$\frac{dP}{d\Omega} = \frac{\epsilon_0 mc^3}{4\pi} \frac{\beta^4}{\rho^2} \frac{(1 - \beta \cos \Theta)^2 - (1 - \beta^2) \sin^2 \Theta \cos^2 \phi}{(1 - \beta \cos \Theta)^5}$$

Integrate: $P_r = \frac{2}{3} \epsilon_0 mc^3 \frac{\beta^4 \gamma^4}{\rho^2}$

Also find that $\frac{dP}{d\Omega} \propto \left(\frac{1}{\gamma^2} + \Theta^2 \right)^{-3}$

→ most of the power radiated
in a cone w. $\Theta_r = 1/\gamma$

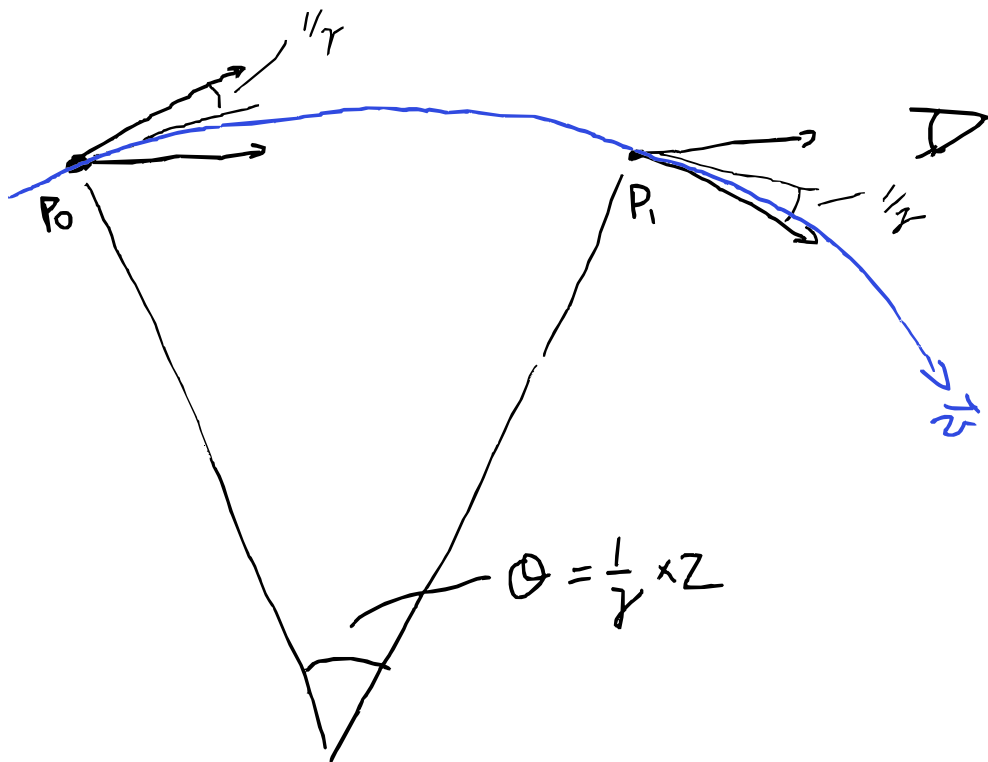


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Radiation Spectrum

Broadband radiation, up to some cut-off.

Let's estimate the time an observer is illuminated by the synch. light:



Path that the photon takes from $P_0 \rightarrow P_1$

$$l_\gamma = Z \rho \sin \frac{\Theta}{2} = Z \rho \sin \frac{1}{\gamma} \quad (\text{chord})$$

$$\rightarrow \boxed{t_\gamma = \frac{Z \rho \sin 1/\gamma}{c}}$$

For the e^- : $l_e = \rho \Theta = \frac{Z \rho}{\gamma} \therefore \boxed{t_e = \frac{Z \rho}{\beta c \gamma}}$ 16

The the amount of time the radiation
flashes the observer is:

$$\begin{aligned}\delta t &= t_e - t_r \\ &= \frac{Z\rho}{B\gamma} - Z\rho \frac{\sin 1/r}{c}\end{aligned}$$

$$\delta t \approx \frac{4\rho}{3c\gamma^3}$$

→ this sets the max freq. of the
light that is emitted.

Define the critical frequency as:

$$\omega_c \approx \frac{1}{\frac{1}{2}\delta t} \approx \frac{3}{2}c \frac{\gamma^3}{\rho}$$

$$\Rightarrow E_c(\text{keV}) = 0.665 E^2[\text{GeV}] B[\text{T}]$$

∴ You need GeV scale beams

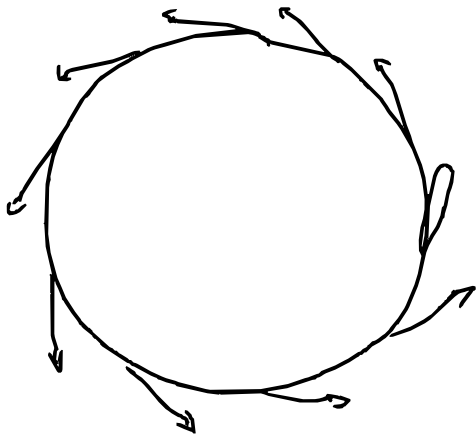
+

T scale magnets

→ keV Xrays.

Energy loss per turn

(See Weidemann 24.2)



e^- loses energy
through synch.
rad around
the ring.

Energy loss per turn is:

$$U_L = \oint P(s) dt = \oint P(s) \frac{ds}{c}$$

Remember: $P = \frac{2}{3} r_e m c^3 \frac{\beta^4 \gamma^4}{\rho^2}$

For isomagnet ring $\rightarrow \rho = \text{const}$

$$\therefore U_L = C_r \frac{E_0^4}{\rho}$$

In engineering units:

$$U_L (\text{keV/turn}) = \frac{88.5 E^4 [\text{GeV}]}{\rho [\text{m}]}$$

$\rightarrow$ Can also turn this into Power loss / turn. ¹⁸